4/1/2022

Thursday, March 31, 2022 11:07 PM

M168_Notes_8.2

8.2 Expected Value

The expected value for the variable X in a probability distribution is

$$\boxed{E(X) = X_1 \cdot P(X_1) + X_2 \cdot P(X_2) + \cdots + X_n \cdot P(X_n)}$$

Example

You are dealt a hand of 5 cards. What is the expected number of hearts?

$E = 0 \times 0.222 + 1 \times 0.232 + 2 \times 0.274 + 3 \times 0.082 + 4 \times 0.011 + 5 \times 0.0005$

≈ 1.305

Example

A sample of mini boxes of raisin bran cereal was selected and the number of raisins in each box was counted. The results are shown in the table below.

# of boxes	13	14	15	16	17
# of raisins	10	14	19	16	12

Total 75

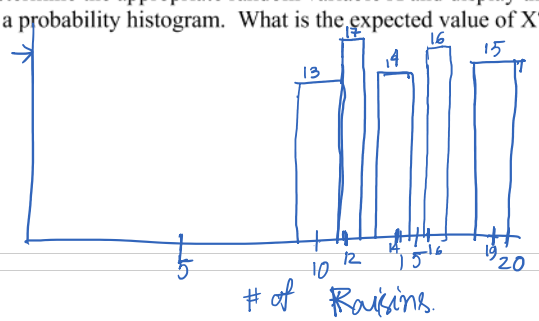
$P(X=10)$

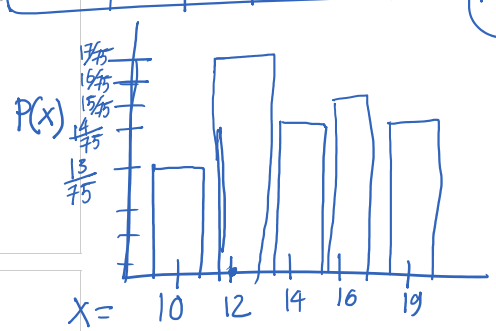
$P(X)=$ Freq. of boxes	$\frac{13}{75}$	$\frac{14}{75}$	$\frac{15}{75}$	$\frac{16}{75}$	$\frac{17}{75}$
$X=$ # of Raisins	10	14	19	16	12

← Frequency table.

10 12 14 16 19

Determine the appropriate random variable X and display the data in a probability histogram. What is the expected value of X?





What is the MEAN of X?

$E(X) = x_1 \cdot P(x_1) + \cdots + x_n \cdot P(x_n)$

$= 10 \cdot \frac{13}{75} + 14 \times \frac{14}{75} + 19 \times \frac{15}{75} + 16 \times \frac{16}{75} + 12 \times \frac{17}{75} = 14.28.$

What was the X value that happened the most often?

12

What was the X value that was in the middle?

14

~~What is the RANGE of X values?~~

Histograms and Averages:

The MEAN (expected value) is where the histogram "balances"

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The MODE is the tallest rectangle.

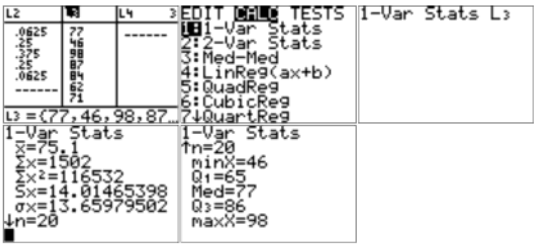
The MEDIAN is where the area is cut in half.

The RANGE is the number of rectangles. (remember, some may have a height of 0).

Example

Find the mean, median and mode of the following test scores:

77, 46, 98, 87, 84, 62, 71, 80, 66, 59,
79, 89, 52, 94, 77, 72, 85, 90, 64, 70



Mean = 75.1 Median = 77

Mode = Range =

Another way to measure spread? QUARTILES

46 52 59 62 64 66 70 71 72 77 77 79 80 84 85 87 89 90 94 98

Q1 = Q3 = IQR =

Box and whisker plot

Example

A game consists of choosing two bills at random from a bag containing 7 one dollar bills and 3 ten dollar bills. The player gets to keep the money picked. How much should be charged to play this game to keep it "fair" (expected value of zero)?

$7 \to \$1$

$3 \to \$10$

	Event	
$X =$	2	$\frac{C(7,2)}{C(10,2)} \approx \frac{21}{45}$
$X =$	11	$\frac{C(7,1)\cdot C(3,1)}{C(10,2)} = \frac{21}{45}$
$X =$	20	$\frac{3}{45}$

$$E(x) = 2\cdot\frac{21}{45} + 11\cdot\frac{21}{45} + 20\cdot\frac{3}{45}$$

$$\approx 7.4$$

P is charge.

$$E(x) - P = 0 \Rightarrow P = E(x).$$

$$P = \$7.4$$

Example

From a group of 2 women and 5 men a delegation of 2 is chosen. Find the expected number of women in the delegation.

ODDS

If P(E) is the probability of event E occurring, then the odds in favor of E are

$$\frac{P(E)}{1-P(E)} = \frac{P(E)}{P(E^c)}, \quad P(E) \neq 1$$

We usually express the odds as a ratio of whole numbers,

$$\frac{a}{b} \quad \rightarrow \quad a \ to \ b \quad \rightarrow a{:}\,b$$

Example

If the probability that the Aggies will win a football game is 80%, what are the odds in favor of the Aggies?

If we are given the odds we want to be able to find the probability, if the odds in favor of E are given as a:b, then

$$P(E) = \frac{a}{a+b}$$

Odds to win the Breeders' Cup Distaff
Saturday, November 4th, 2006

Siempre 29/5	Balletto 7/5	Bushfire 18/1
Fleet Indian 2/1	Happy Ticket 21/2	
Round Pond 22/1	Spun Sugar 11/1	Summerly30/1